

Regioselective Fluorohydrin Synthesis from Allylsilanes and Evidence for a Silicon-Fluorine Gauche Effect

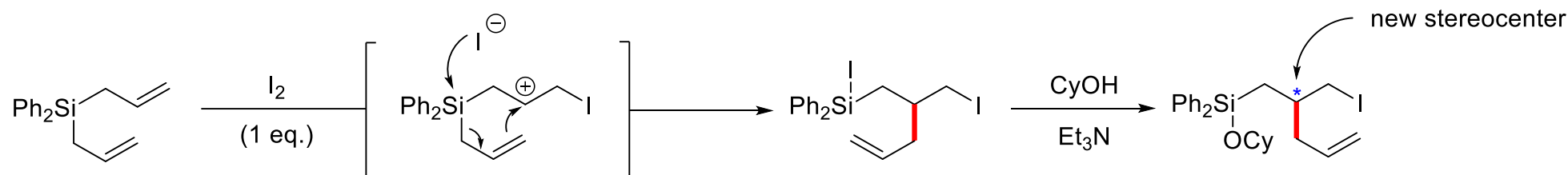
Thesis Defense

Alexie Clover

5/24/2023

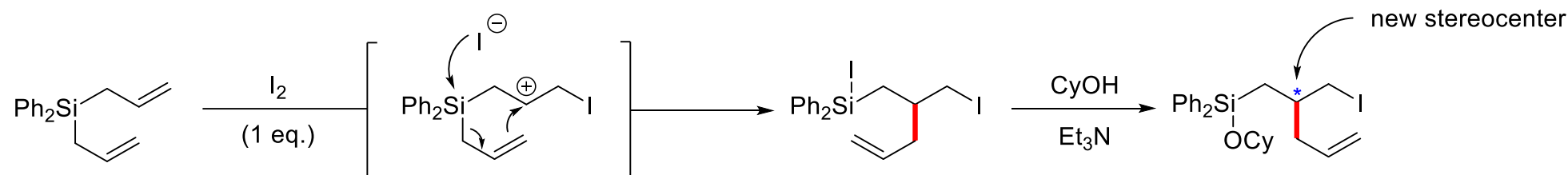
This thesis will focus on further developing both silicon-tethered reactions discovered in the O'Neil lab with an eye toward generating useful intermediates that can be used to synthesize biologically active small molecules.

1. Starting from the iodine mediated rearrangement product one can synthesize increasingly complex structures that contain more reactive sites.

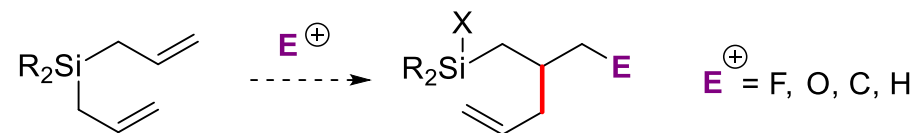


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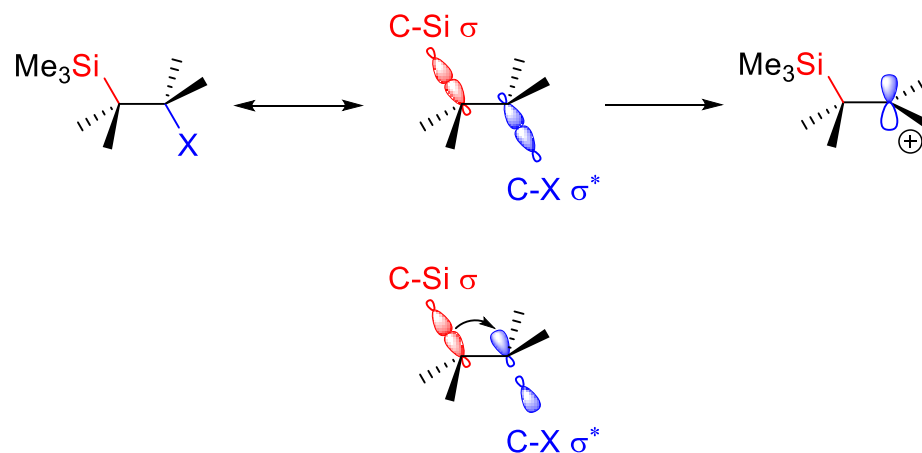
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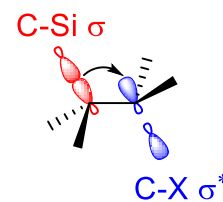
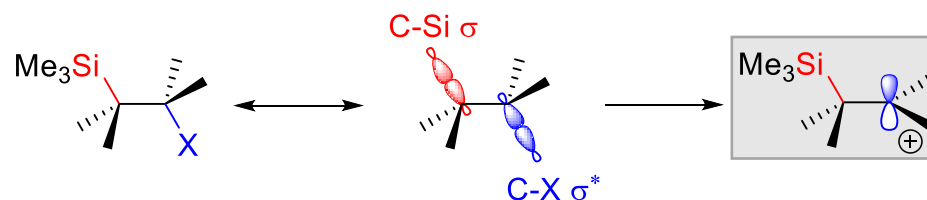
2. One aim is to expand the diallylsilane rearrangements to include elements other than iodine. In particular, the development of an epoxide-promoted rearrangement is of significant interest due to the value of further modification.



- The β -silyl effect is a type of hyperconjugation where silicon stabilizes a positive charge on a carbon in the β -position.
- This occurs by the C-Si σ orbital donating electron density into the empty p-orbital in the β position to stabilize the resulting carbocation.
- The C-Si σ orbital can donate electron density into the C-X σ^* orbital, thus weakening the bond.

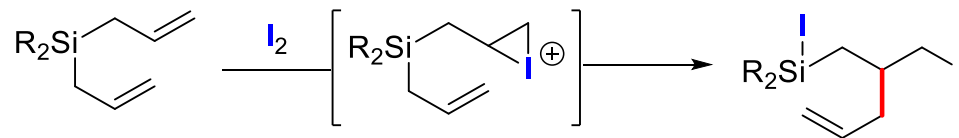


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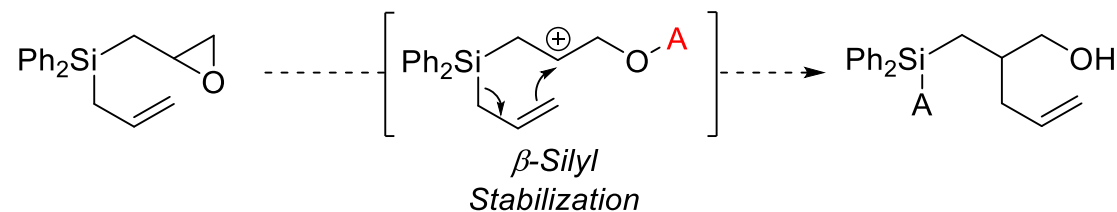
We'll come back to this later

Previously:

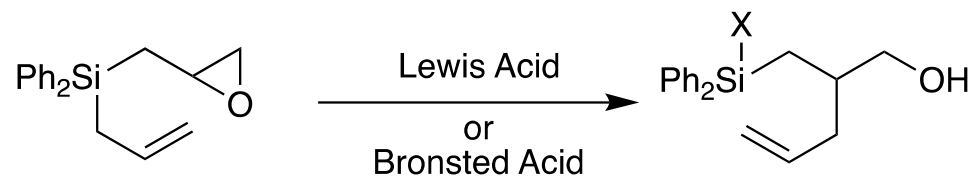


Cummins and O'Neil, *Tet. Lett.* **2017**, 58, 3406

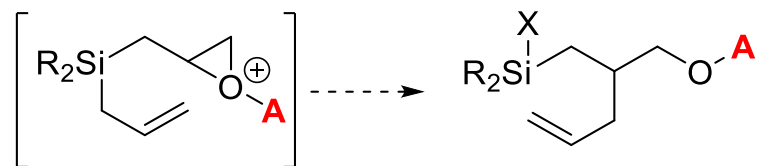
Proposed intramolecular epoxide allylation:



With this proposed rearrangement in mind we decided to start screening.

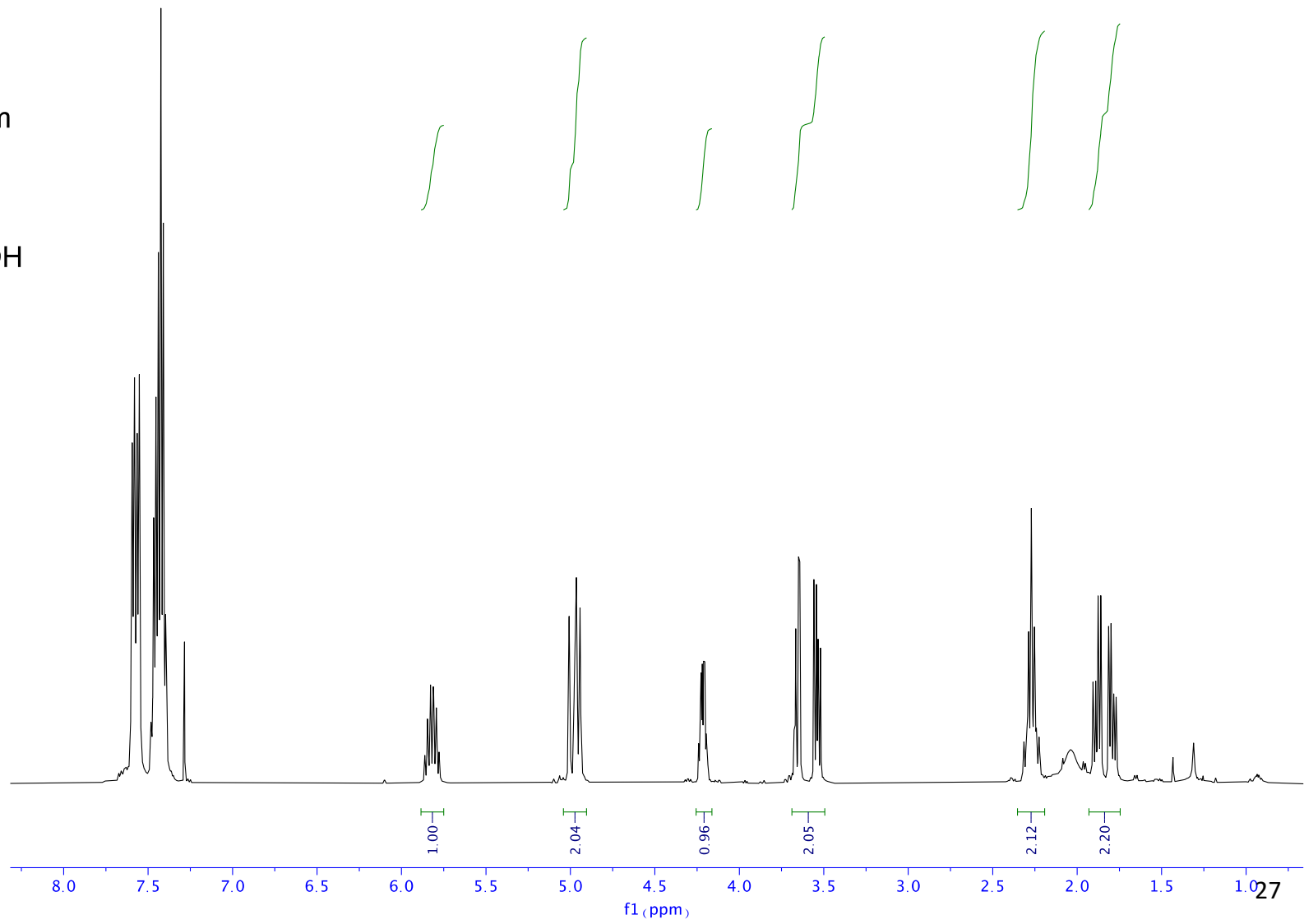
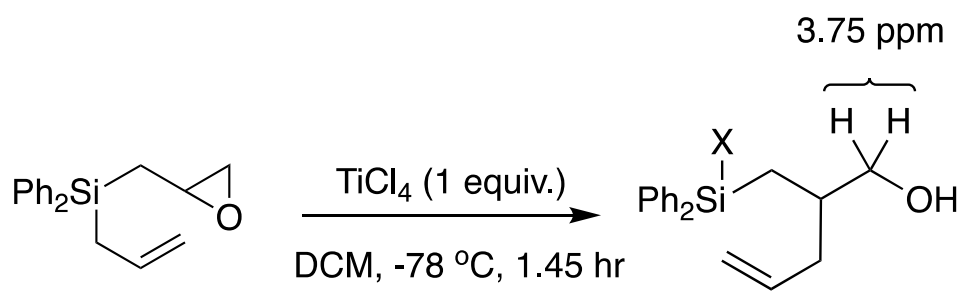


via



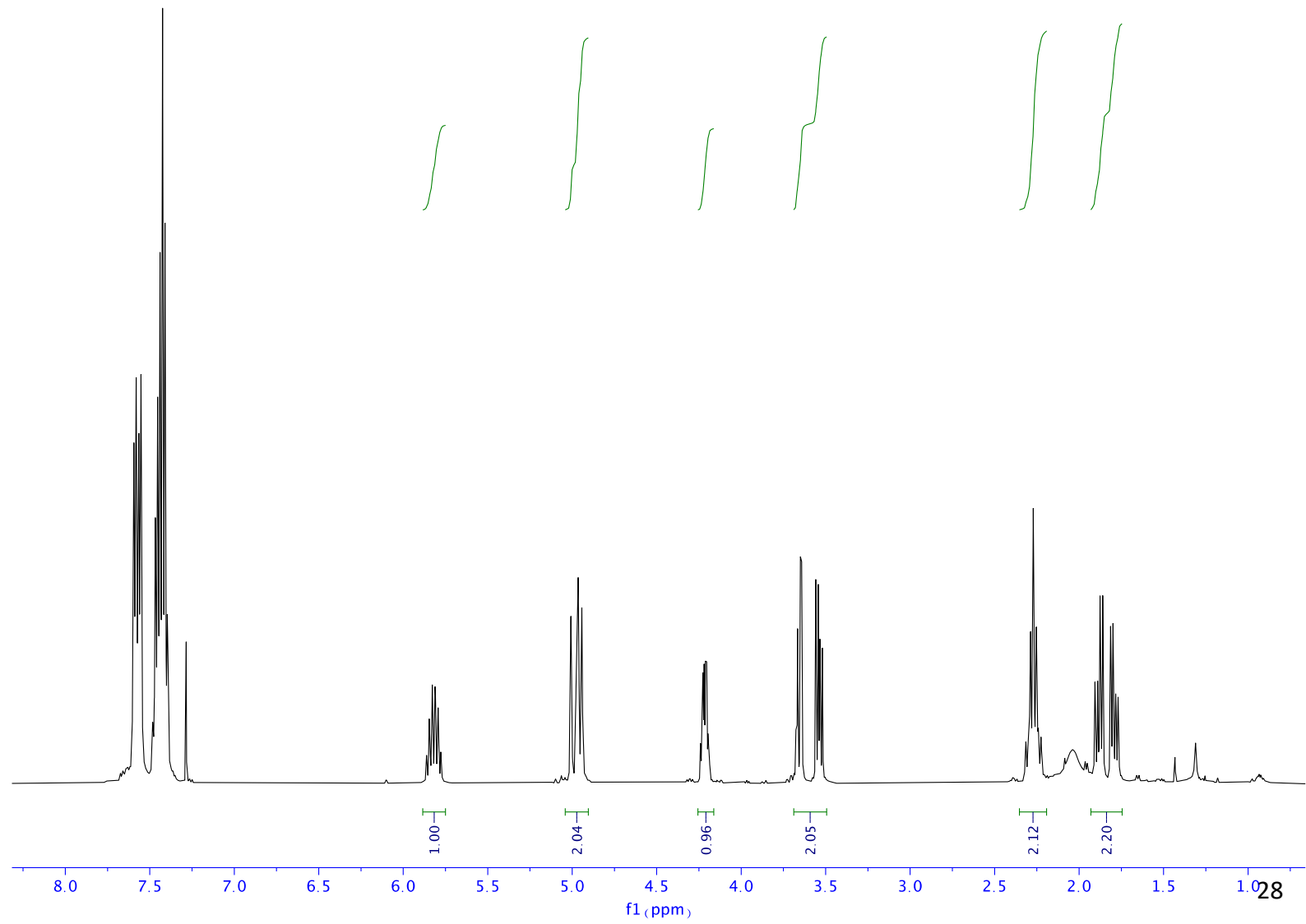
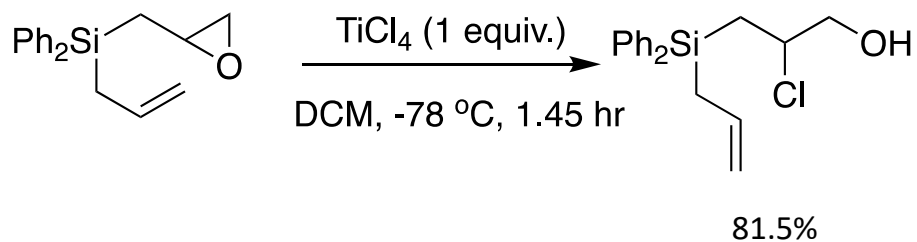
Rearrangement with the epoxide

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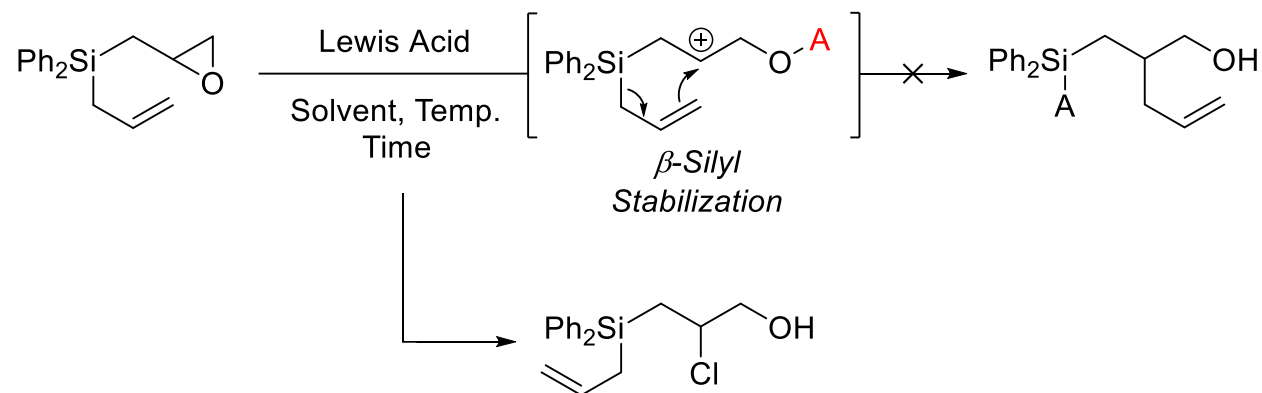


Rearrangement with the epoxide

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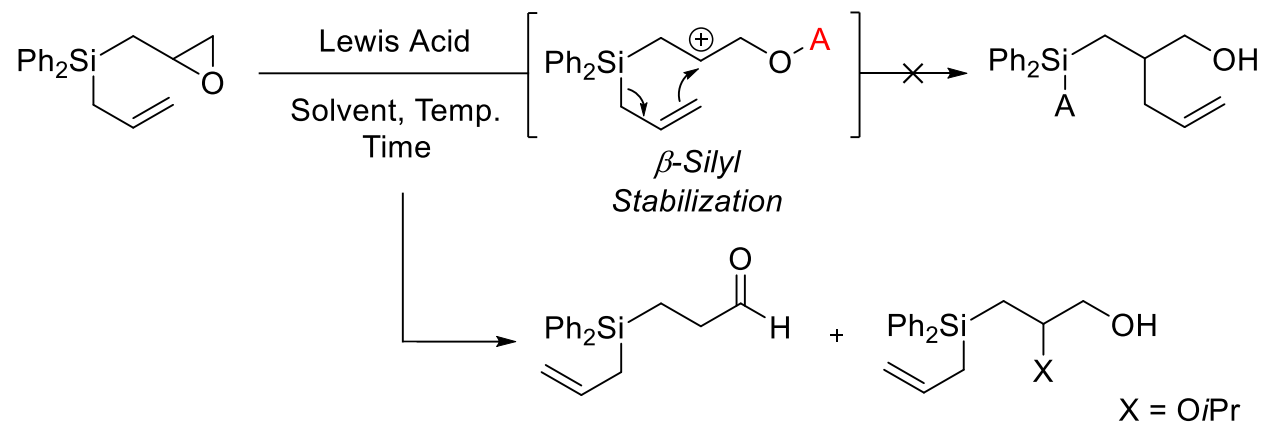
Attempts at an Intramolecular Epoxide Allylation



Entry	Lewis Acid	Solvent	Temp. (°C)	Time (hr)	Conv. %
1	TiCl ₄	DCM	-78	1	44
2	TiCl ₄	DCM	-78	2	82 ^b
3	TiCl ₄	MeCN	-20	2	100
4	(Et) ₂ AlCl	DCM	-78	1	100
5	(Et) ₂ AlCl	DCM	25	0.5	100
6	(Et) ₂ AlCl	DCM	25	1	100
7	Me ₃ SiCl	DCM	-78	1	~50
8	Me ₃ SiCl	DCM	25	1	100

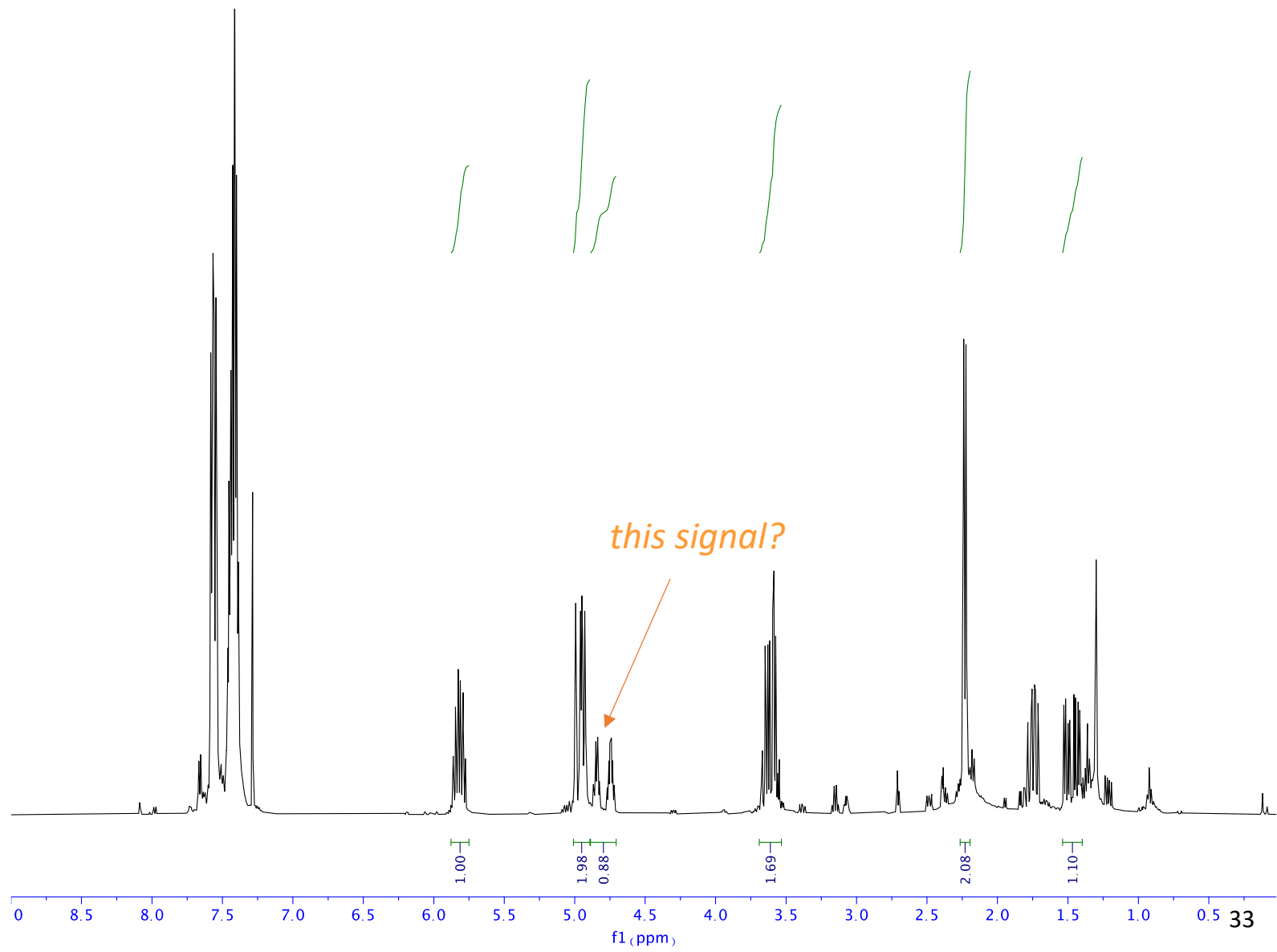
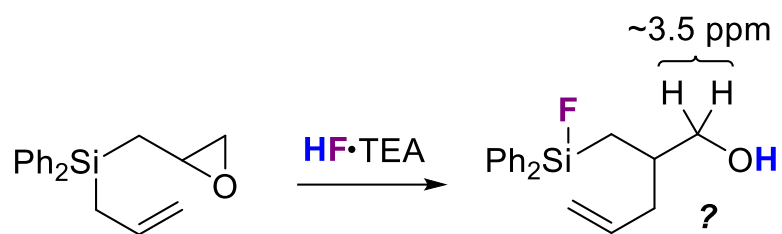
^aConv % based on the amount of starting material reacted.

Attempts at an Intramolecular Epoxide Allylation

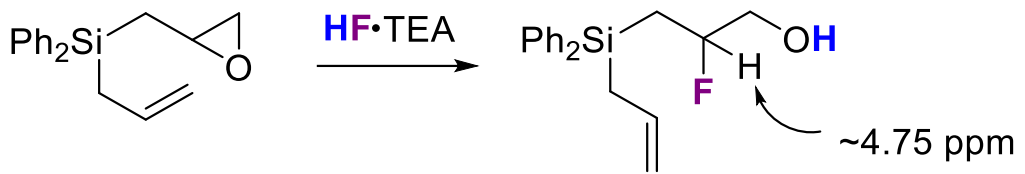


Entry	Lewis Acid	Solvent	Temp. (°C)	Time (hr)	Conv. %
1	BF ₃ •Et ₂ O	DCM	0	1	100
2	BF ₃ •Et ₂ O	DCM	-78	1	100
3	BF ₃ •Et ₂ O	DCM	-78	2	100
4	Ti(O-iPr) ₄	DCM	0	1	0
5	Ti(O-iPr) ₄	DCM	25	16	0
6	Ti(O-iPr) ₄	PhMe	60	4	0
7	Ti(O-iPr) ₄	HFIP	25	4	100
8	TBDMS Triflate	DCM	25	16	100
9	Sn(OTf) ₂	DCM	25	16	100
10	Sc(OTf) ₃	DCM	25	16	100
11	Bi(OTf) ₃	DCM	25	16	100
12	BiBr ₃	DCM	25	16	100

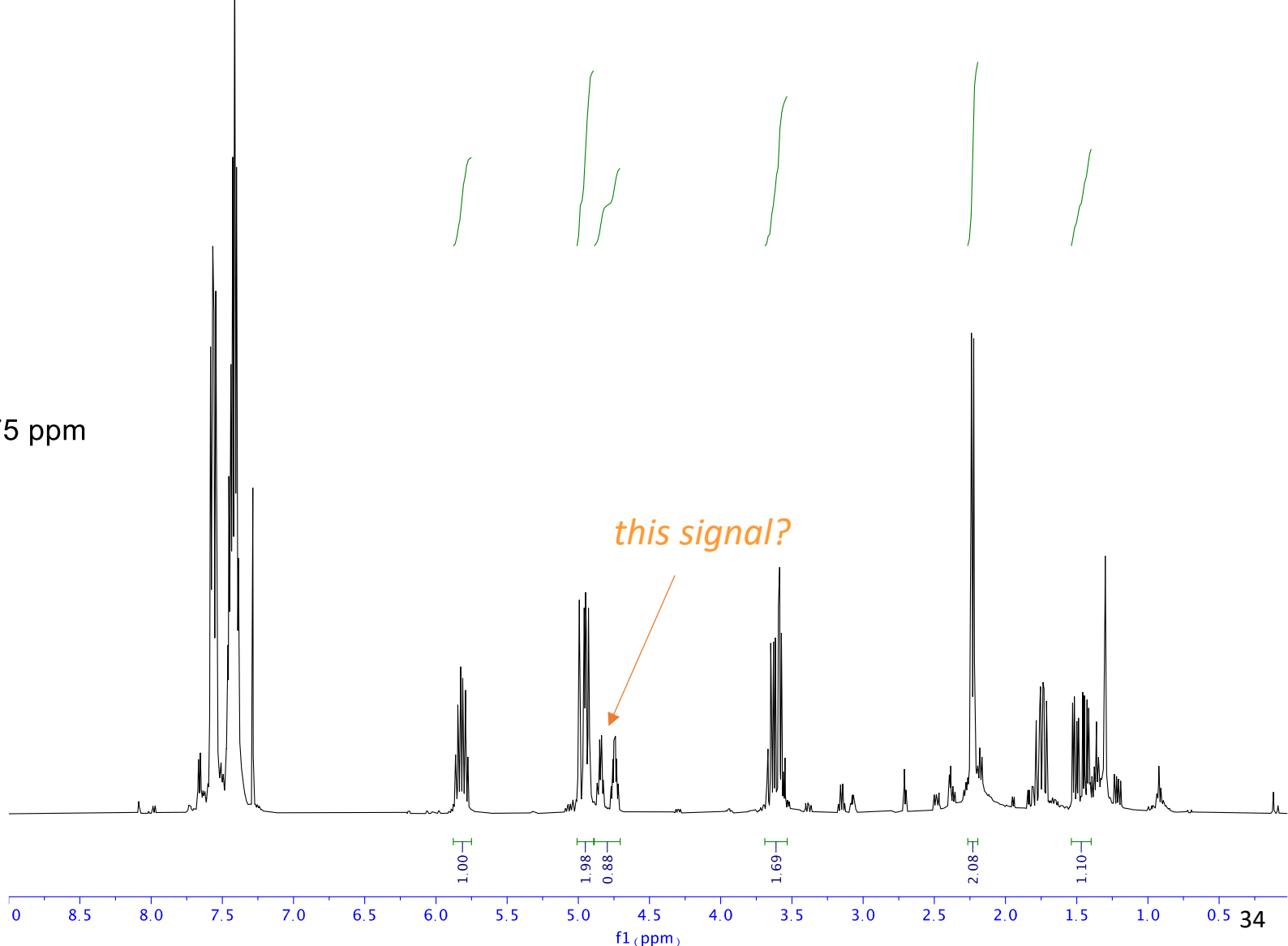
^aConv % based on the amount of starting material reacted.



Use of a Brønsted Acid



JSA-JA-51 (epoxide + HF%3AEt3N).1.f1f
5 eq HF (37% HF:Et3N) with monoepoxide



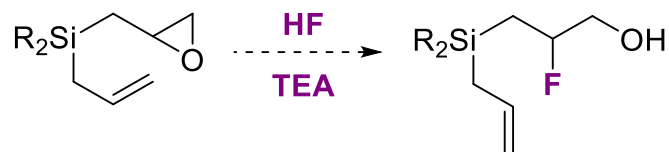
A strive for the addition of fluorine to organic molecules has risen and is in more than 30 % of the 100 best-selling small molecule drugs.

Fluorine has some unique properties including increasing the lipophilicity and potency of the drug.

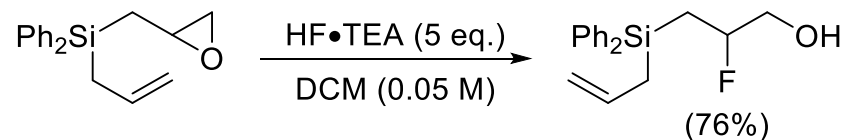
Similar in size to hydrogen but more polar.

Positron-emission-tomography (PET) scans. Labeled fluorine-18 can be used to track the physiology of drugs. Important due to nuclear decay properties of fluorine. The half-life is “just right”.

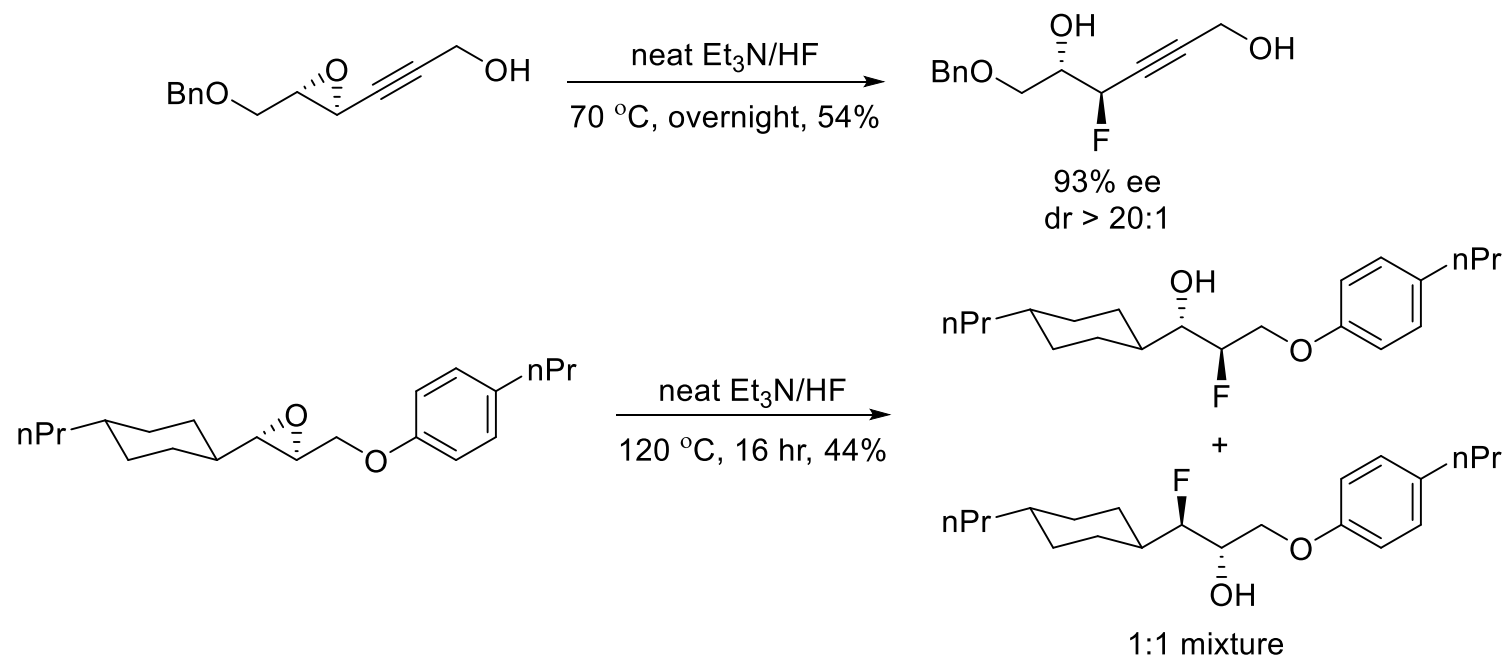
As of August 2021, there are 16 FDA-approved PET tracers, in which 10 of them contain fluorine.



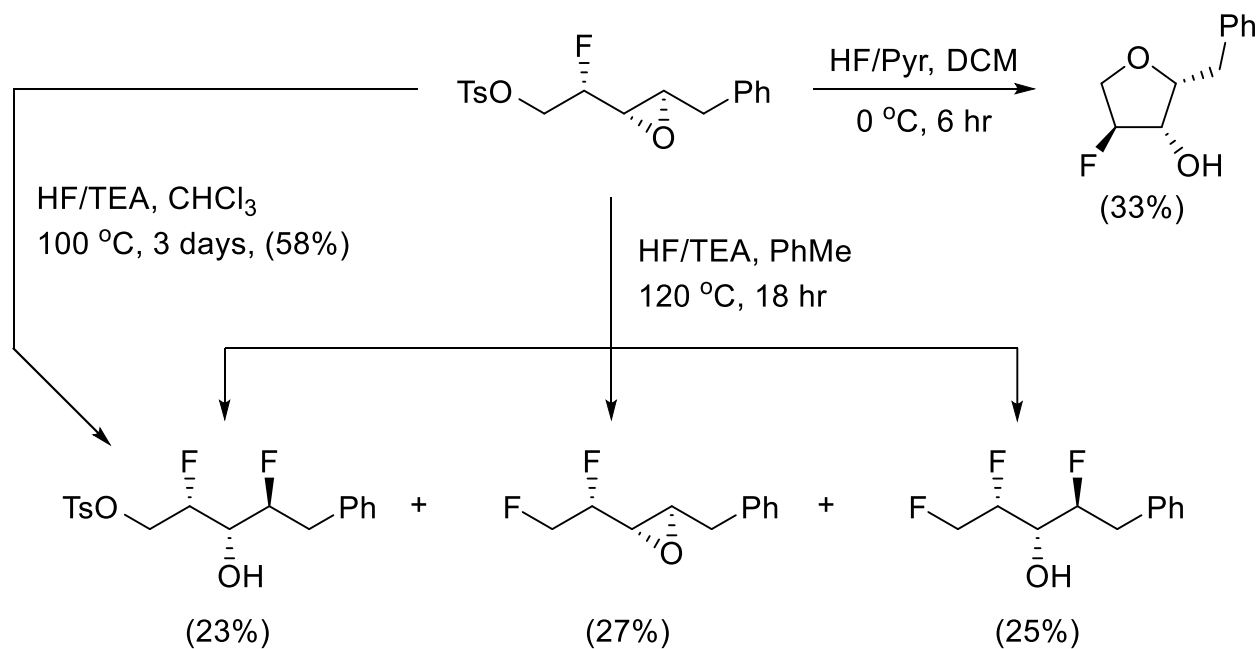
- HF•Pyr led to a more complex mixture.
- HF•TEA was rather clean, with little to no side products.
- The formation of the fluorohydrin took place in mild conditions being that it proceeded at room temperature with five equivalents of HF•TEA.
- This was appealing because the reagent HF•TEA is a milder version, being significantly less acidic and therefore easier to handle.



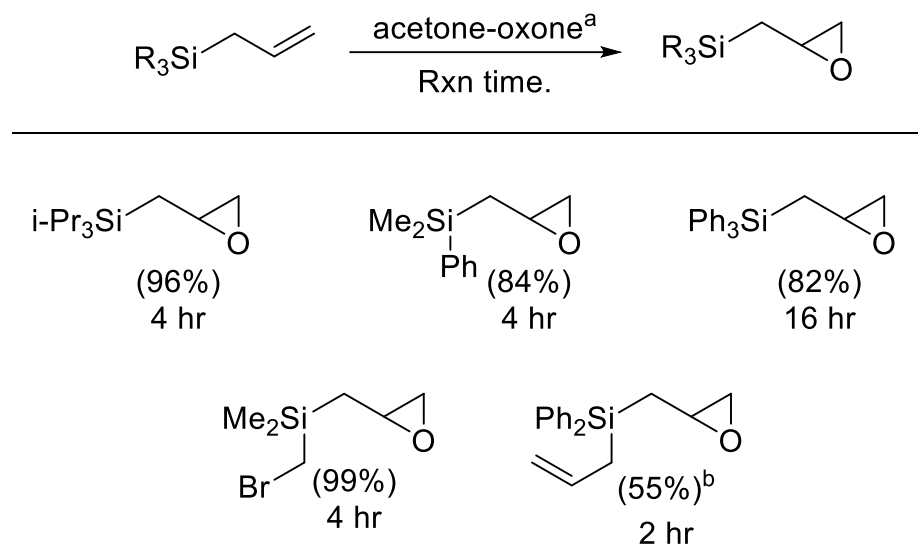
- Usually the conditions need to open epoxides with HF·TEA require long run times with high temperatures.



- Another case of high temperatures with long run times.



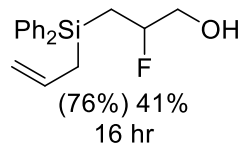
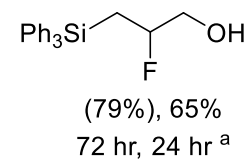
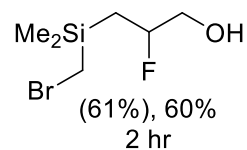
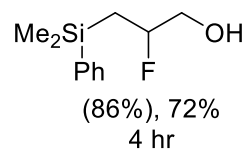
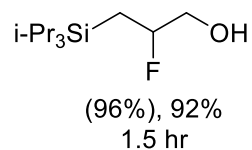
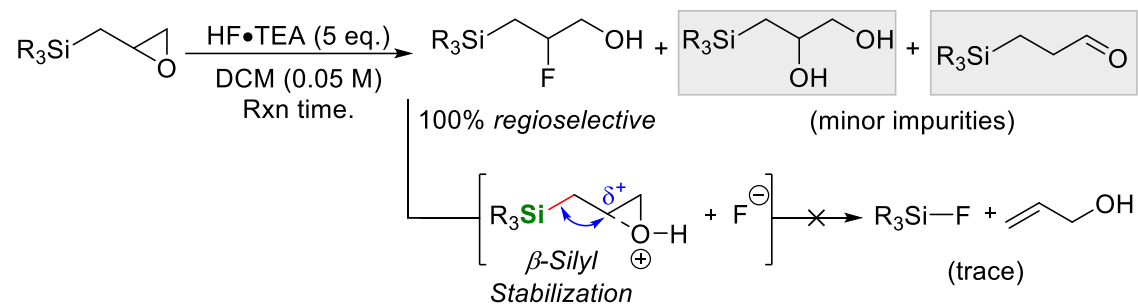
- Exploring other allylsilanes that could be converted to the fluorohydrin.



^a Reagents and conditions: TBAHS (0.04 eq.), K₂CO₃ (0.1 M aq., 0.2 eq.), CH₃CN:DMM (2:1) (0.125 M), oxone (3 eq. in 0.125M 4 x 10⁻⁴ M EDTA), K₂CO₃ (13.3 eq. in 0.125 M H₂O). ^b mCPBA (1.2 eq.), DCM (0.067 M)

Fluorohydrin Synthesis from Epoxides

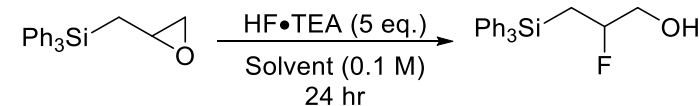
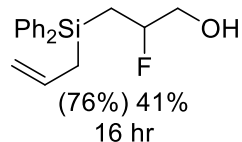
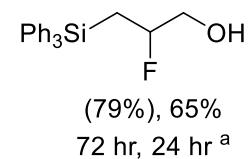
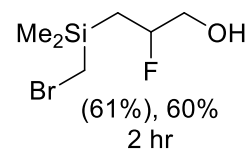
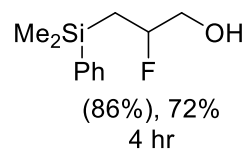
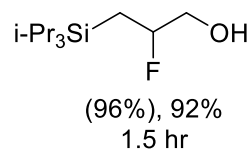
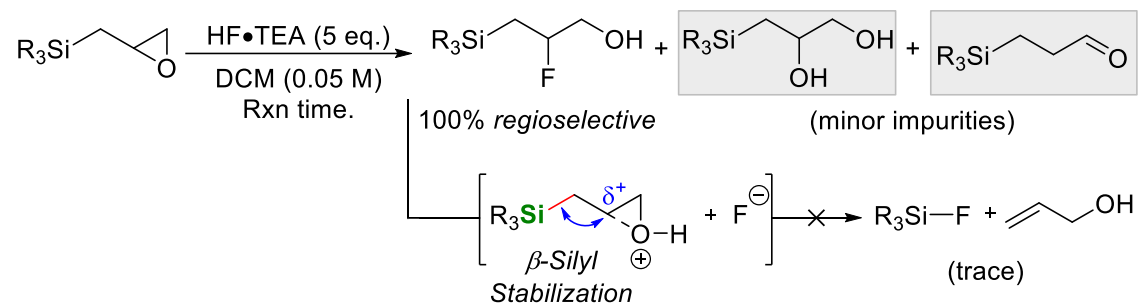
- Transforming epoxysilanes into fluorohydrins under mild conditions.
- Minor impurities include the diol and a Meinwald-type rearrangement.
- Regioselectivity.



The isolated yield is given in parentheses, followed by the overall yield from allylsilanes. ^a HF•TEA (10 eq.), DCM (0.1 M).

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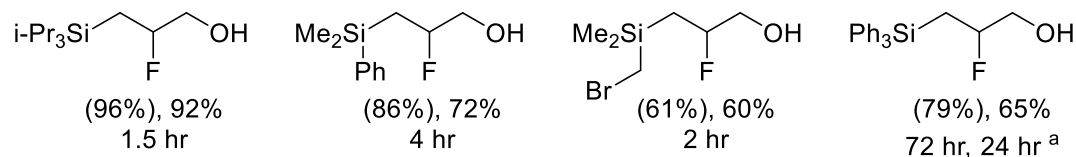
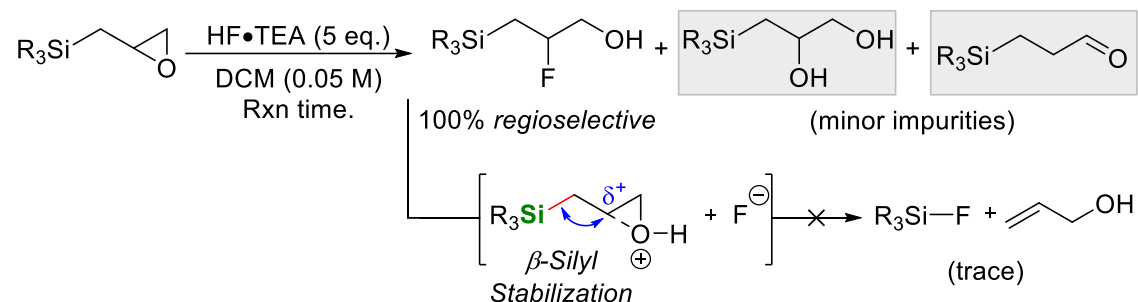


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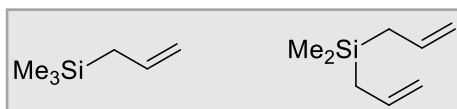
Entry	Solvent	Conv. % to final prod.
1	DCM	50
2	MeCN	18
3	THF	0

Fluorohydrin Synthesis from Epoxides

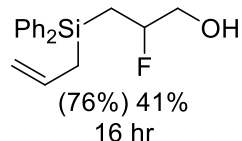
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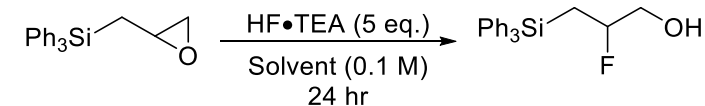
Missing allyl TMS and diallyldimethyl



(Volatility issues)



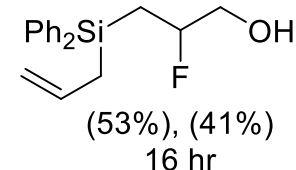
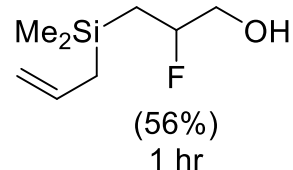
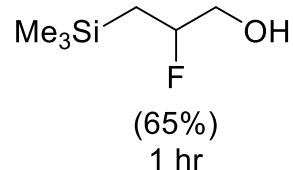
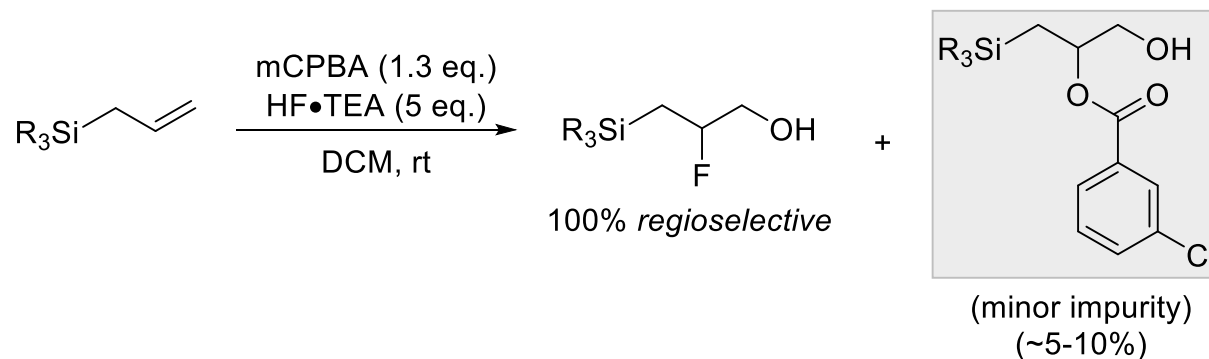
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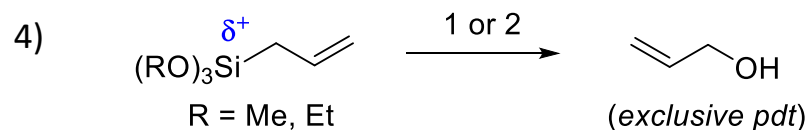
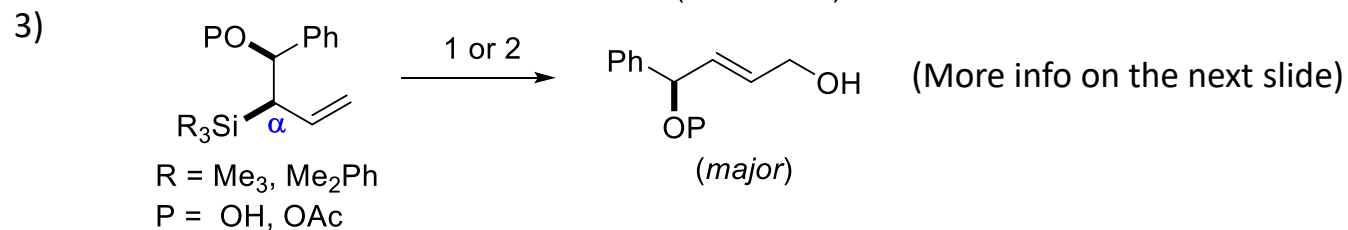
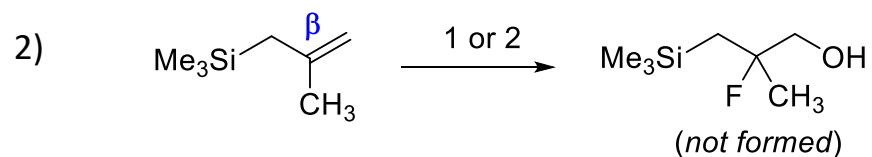
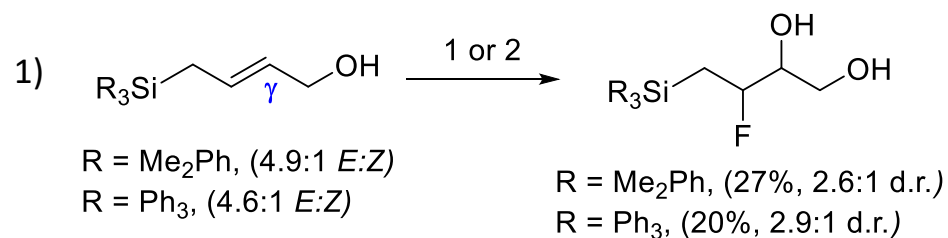
Entry	Solvent	Conv. % to final prod.
1	DCM	50
2	MeCN	18
3	THF	0

One-pot Synthesis of Fluorohydrins

- To our satisfaction, the TMS fluorohydrin was isolated along with the allyldimethylsilyl fluorohydrin and allyldiphenylsilyl fluorohydrin in higher yield.
- Formation of hydroxy ester (minor impurity).

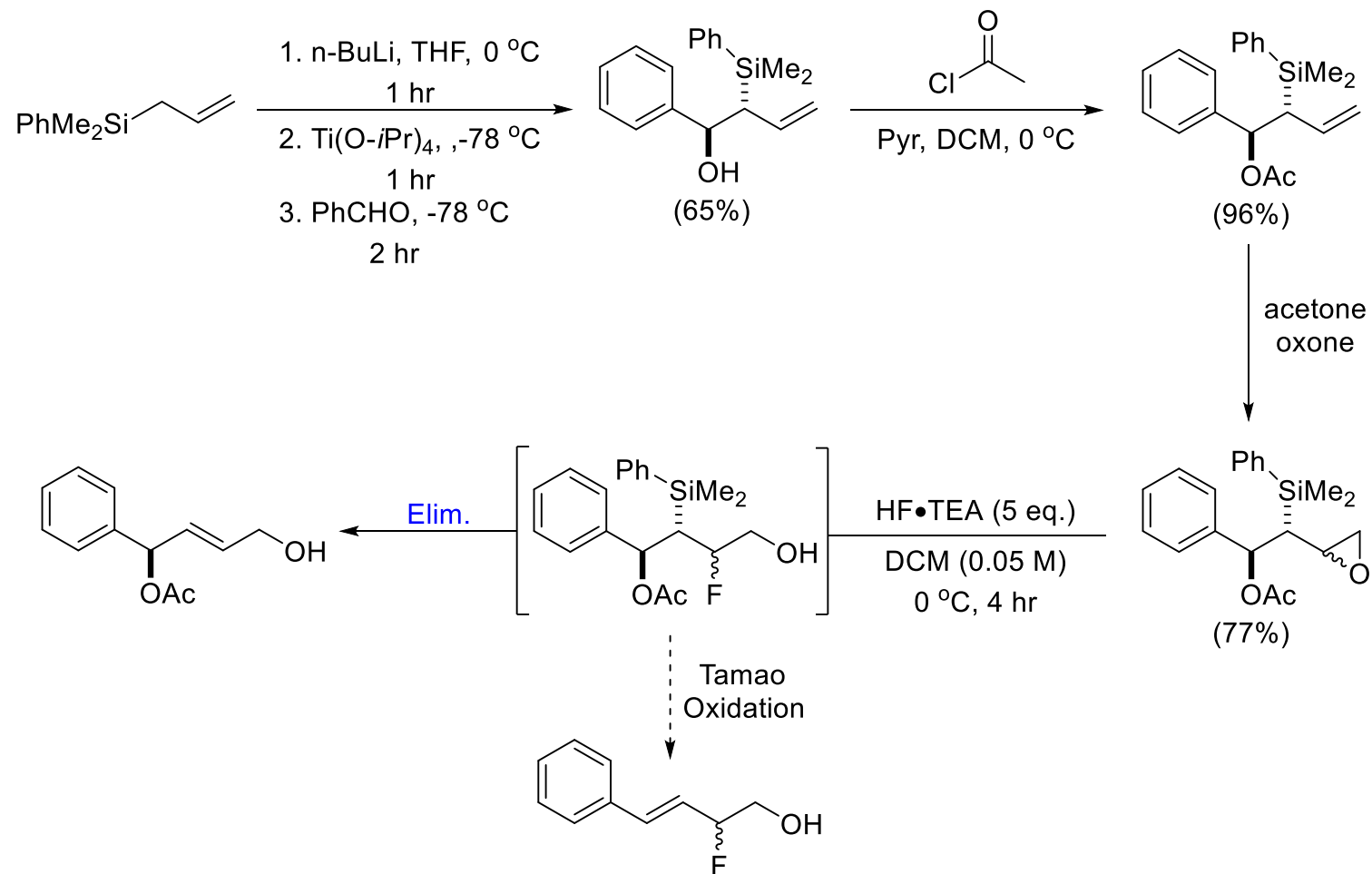


- Attempts at an alpha, beta, gamma substituted allylsilane (ex: 1-3)
- Attempting a more “charged” allylsilane (ex: 4)

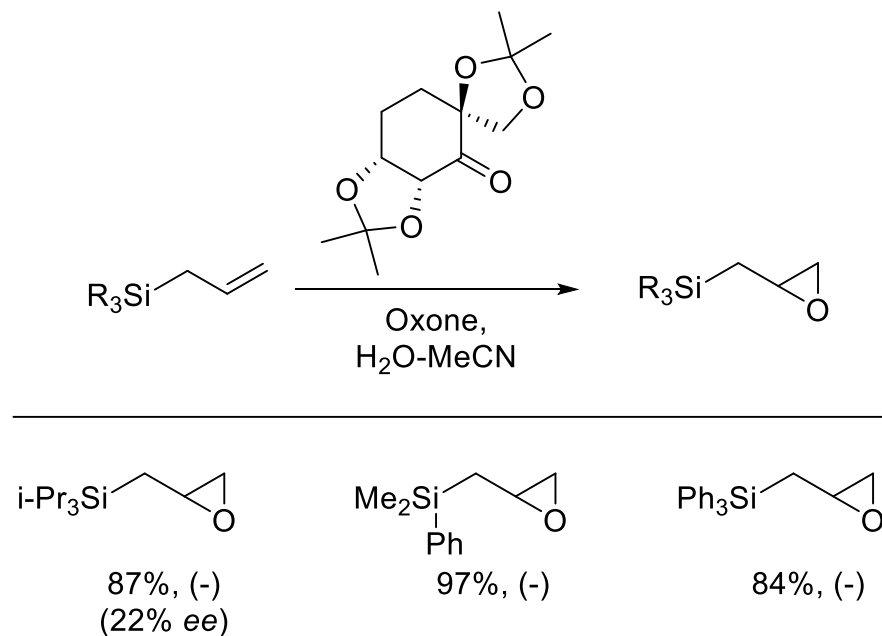


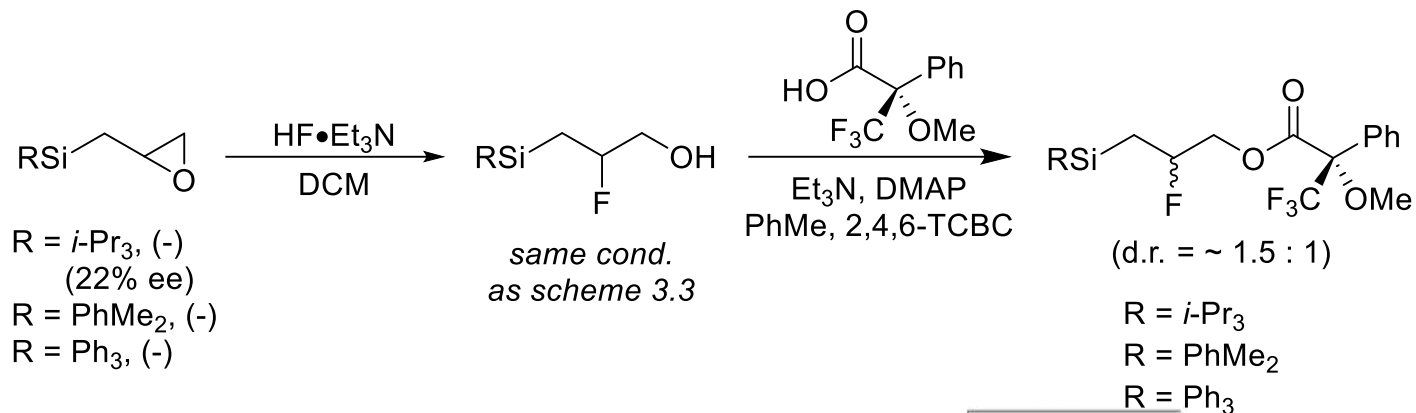
Reagents and conditions: (1) mCPBA, HF•Et₃N, DCM, rt. (2) (a) TBAHS, K₂CO₃, oxone, acetone, DMM-MeCN. (b) HF•Et₃N, DCM, rt.

- We hoped using a less sensitive allylsilane would allow us to make this transformation

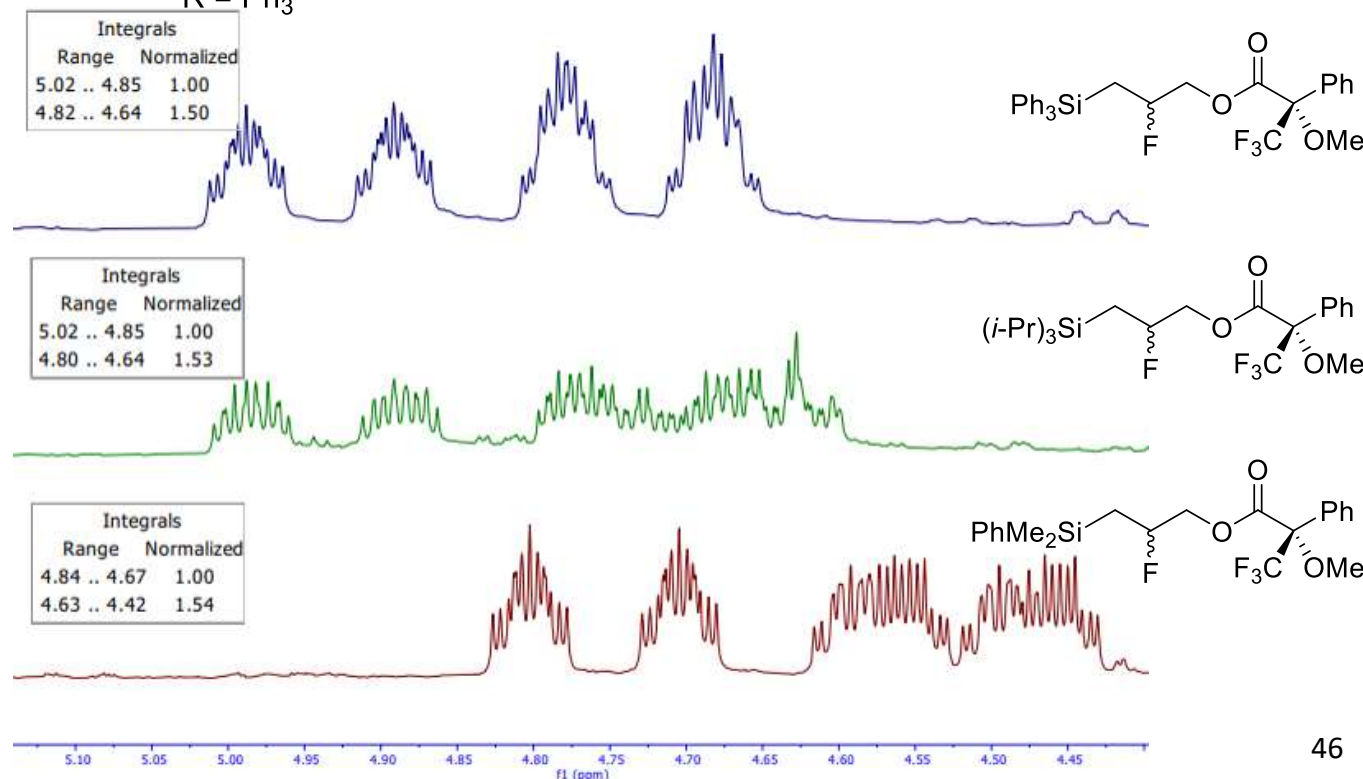


- To determine if the mechanism of this transformation was more S_N1 or S_N2 like, three different enantioenriched epoxides were prepared.
- Using the measured optical activity for triisopropyl silyl epoxide, a comparison was made with what was reported previously in literature.



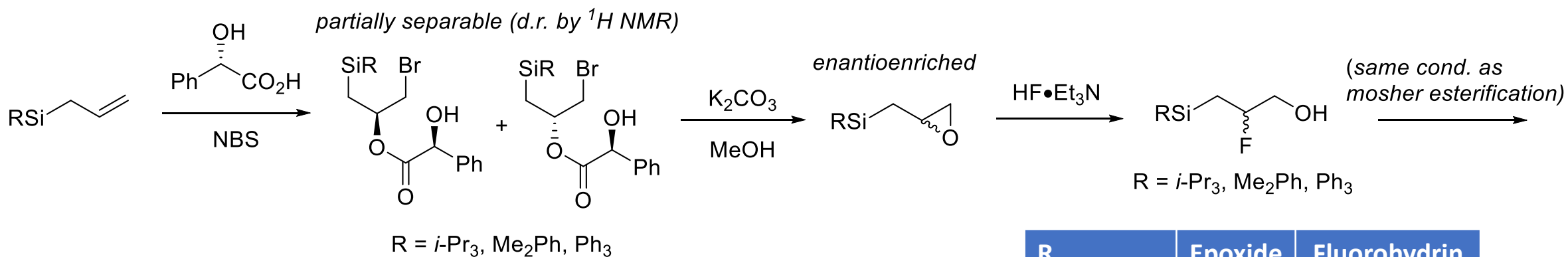


- Subjecting the enantioenriched epoxides to HF then followed by esterification from the chiral carboxylic acid.
- The epoxide *ee* and fluoroester d.r. were similar, suggesting S_N2.



Probing the Mechanism

- Complete retention of *ee* from the *i*-Pr₃. However, an erosion of *ee* was observed for the other two epoxysilanes.
- Suggests S_N2 type mechanism.

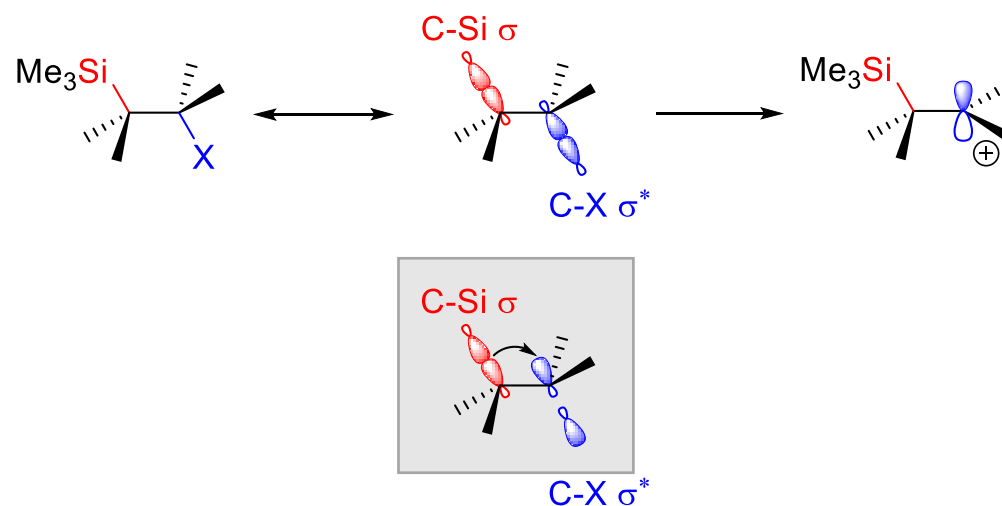


R	Epoxide ee ^A	Fluorohydrin ee ^B
<i>i</i> -Pr ₃	60	60
Me ₂ Ph	64	38
Ph ₃	59	43

Notes: ^Aee from d.r. of bromomandelation.

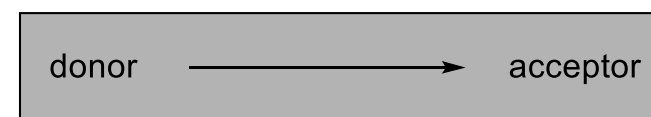
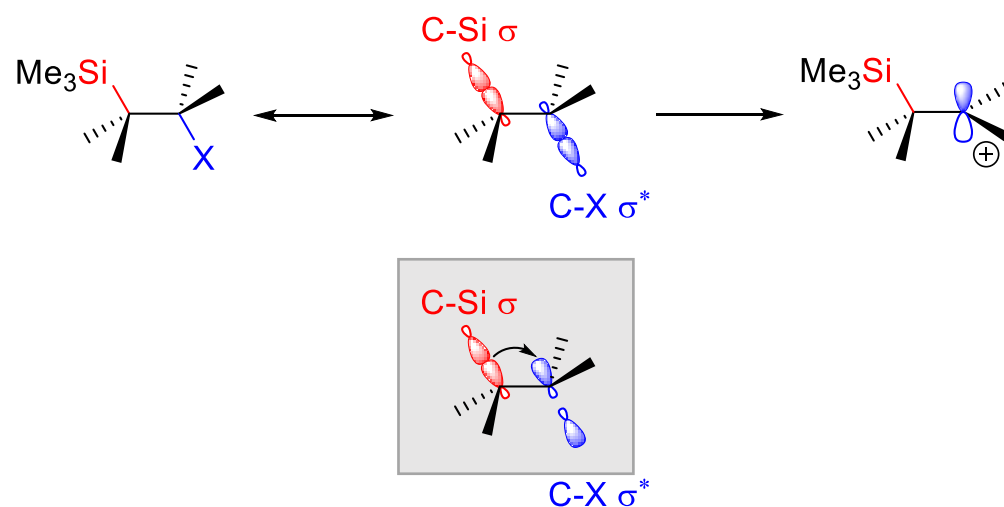
^Bee from d.r. of Mosher esterification.

- The C-Si σ orbital can donate electron density into the C-X σ^* orbital, thus weakening the bond.
- This orbital interaction is maximized in the *anti* conformation.
- C-H σ orbital can donate electron density into the C-F σ^* orbital stabilizing this effect.
- Reported with fluorine, oxygen, nitrogen, and most recently sulfur.

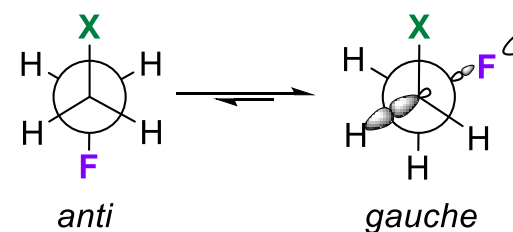
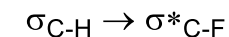


The Beta-Silicon Effect / Fluorine *gauche* Effect

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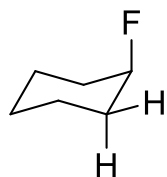
gauche effect



$\text{X} = \text{F}, \text{O}, \text{N}, \text{S}$

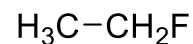
- Hyperconjugation considerations would suggest a preferred anti-periplanar C-Si and C-F arrangement, with stabilization afforded by the overlap between the high energy σ C-Si and the low energy σ^* C-F.
- Interestingly, ^1H NMR analysis provides evidence that the observed conformation is *gauche*.
- Both fluorohydrins display one large (33-35 Hz) and one small (13.5-13.6 Hz) $^3J_{\text{HF}}$ coupling constant.

(A)



$$^3J_{\text{HF}}(\text{anti}, \sim 180^\circ) = 44 \text{ Hz}$$

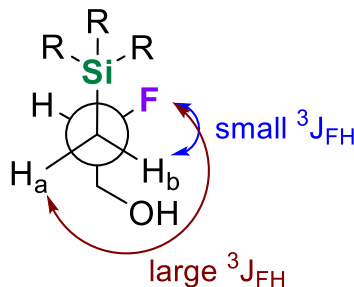
$$^3J_{\text{HF}}(\text{gauche}, \sim 60^\circ) = 10 \text{ Hz}$$



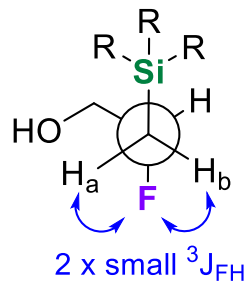
$$^3J_{\text{HF}} = 26.4 \text{ Hz}$$

(Average of the one *anti* and two *gauche* F-H couplings)

(B)



observed

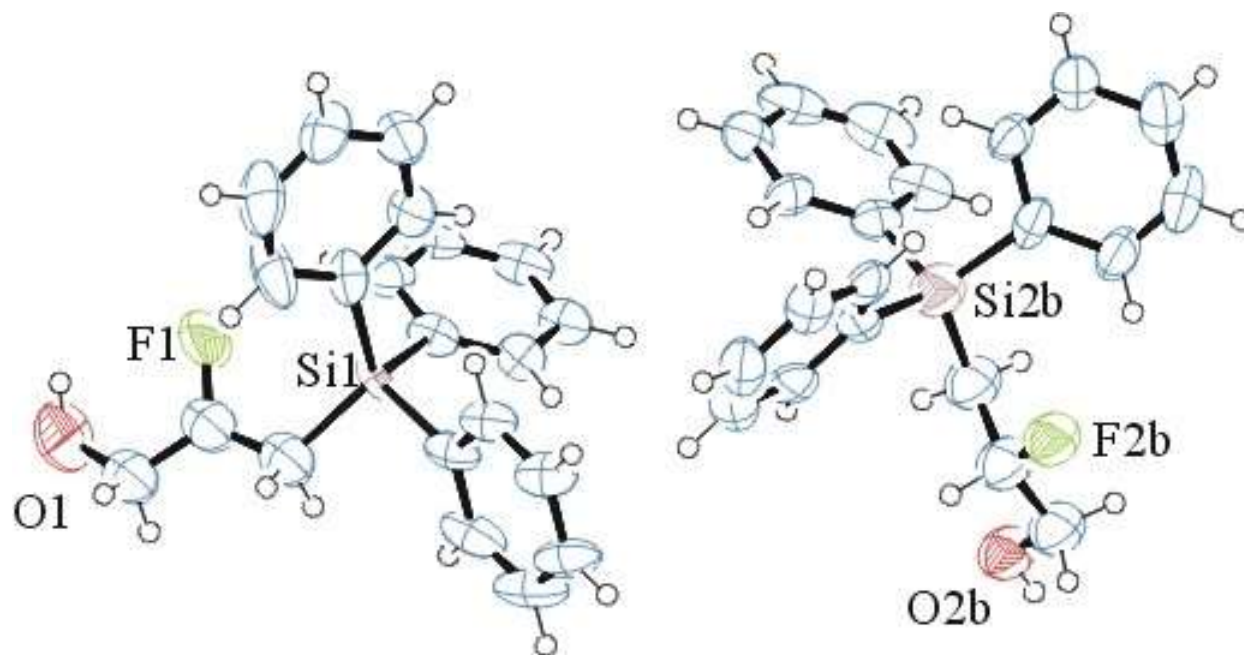


not observed

R	$J_{\text{F-Ha}}$ (Hz)	$J_{\text{F-Hb}}$ (Hz)
Me	32.6	13.6
Ph	35.3	13.5

Crystal Structure of Triphenylsilyl Fluorohydrin

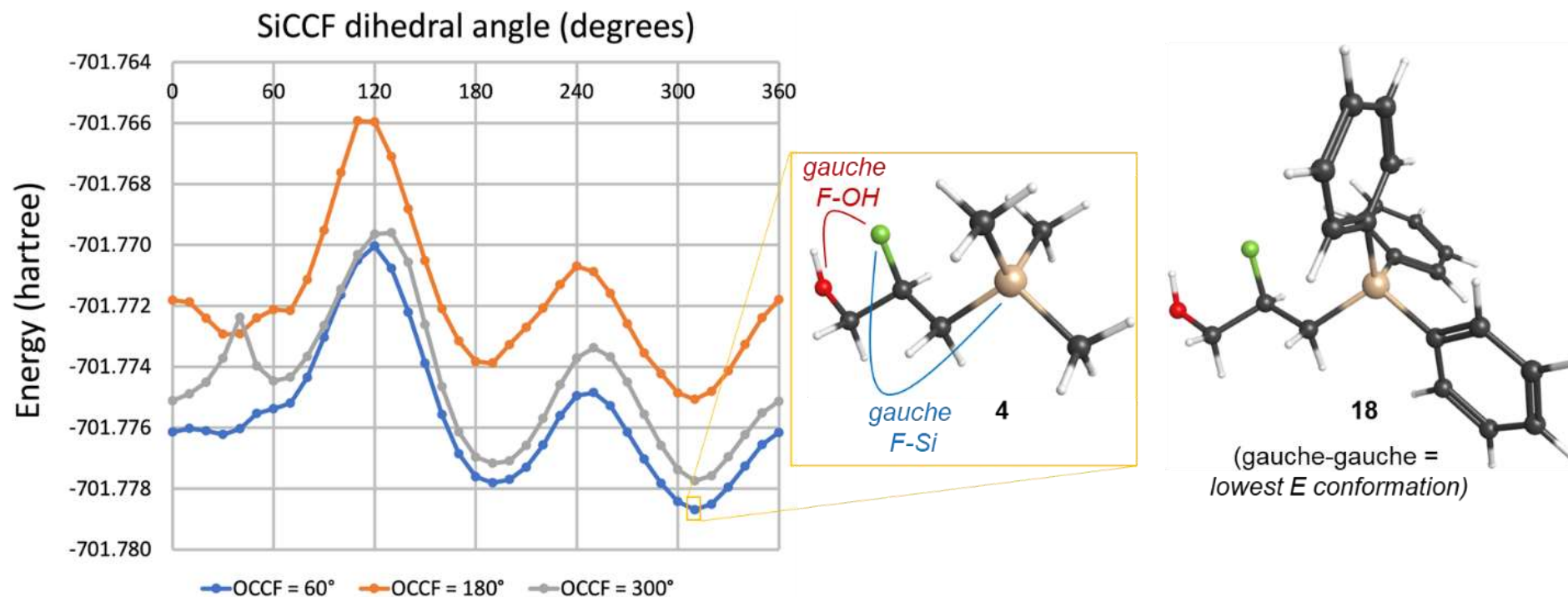
- Two independent structures were observed having different orientations.
- Crystal structure shows *gauche* conformation for both species identified in the unit cell.
- The electron-poor silicon could prove to be electrostatically attractive toward the electronegative fluorine. This would also result in a smaller dipole moment further stabilizing this conformation.



Note the F-OH orientation
($\angle_{\text{F-OH}} = 18^\circ, 179^\circ$)

The dihedral angle between silicon and fluorine was observed to be 6° in the left crystal structure and 38° in the right.

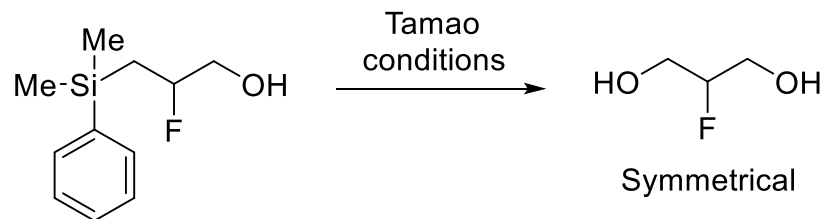
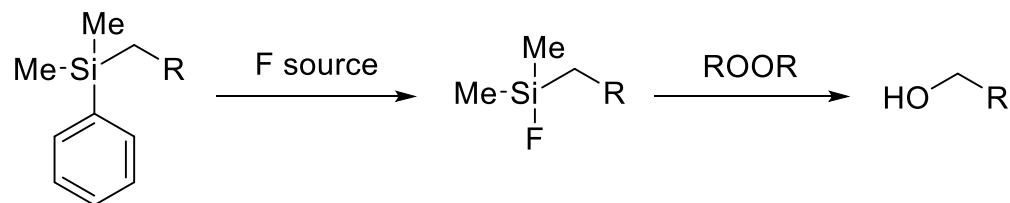
- To better understand this apparent silicon-fluorine gauche effect and its connection to the F-OH orientation in these molecules, conformational analysis by DFT was performed.
- Using both PBE and B3LYP methods, the lowest energy conformation for the trimethyl system **4** have the Si and F groups approximately *gauche* (Si-C-C-F $\phi \sim 310^\circ$) regardless of F-OH orientation ($\phi_{\text{F-OH}} = 60^\circ, 180^\circ, 300^\circ$).
- The absolute energy minimum had both Si-F and F-OH groups *gauche*, which is also true for the Triphenyl system **18** (calculated energy minimum at $\phi_{\text{Si-F}} \sim 50^\circ$ for $\phi_{\text{F-OH}} = 60^\circ$). which is in fairly good agreement with results from X-ray analysis (e.g. $\phi_{\text{Si-F}} = 38^\circ$)



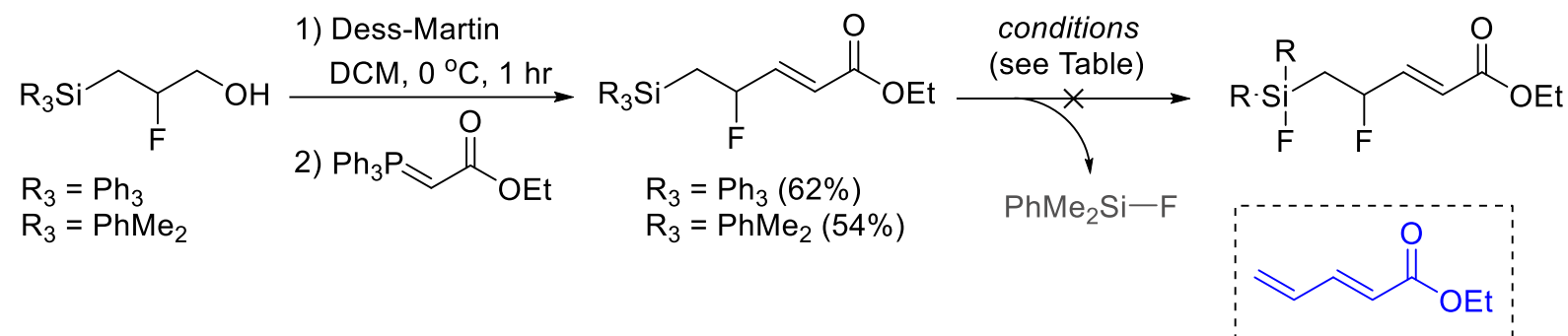
Results from DFT-PBE conformational analysis of 2-fluoro-3-(trimethylsilyl)propan-1-ol (**4**). For three different F-OH orientations ($\phi_{\text{F-OH}} = 60^\circ, 180^\circ, \text{ or } 300^\circ$), the lowest energy conformer contained a *gauche* Si-F arrangement ($\phi_{\text{Si-F}} \sim 310^\circ$).

- The Tamao oxidation converts a carbon-silicon bond to a carbon-oxygen bond.
- This is typically done through the activation of silicon using a fluorine source.
- Once activated, it can be oxidized using peroxy acids or hydrogen peroxide.

General Tamao

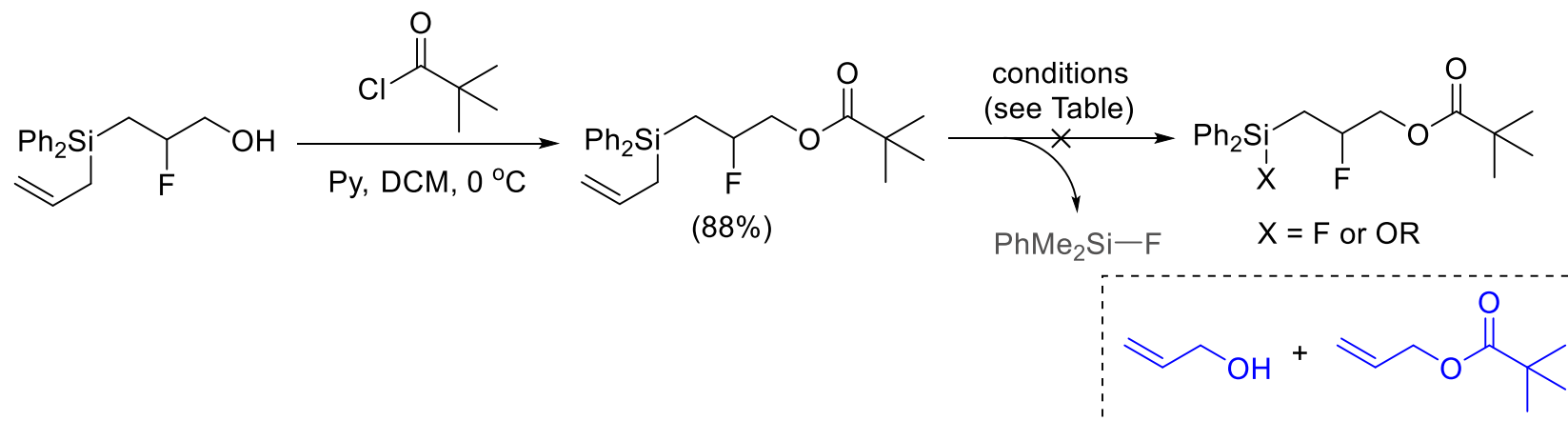


- While silicon is an important element to many functional materials, its conversion to a hydroxy group via a Tamao-Fleming oxidation would lead to a more versatile alcohol functional group.



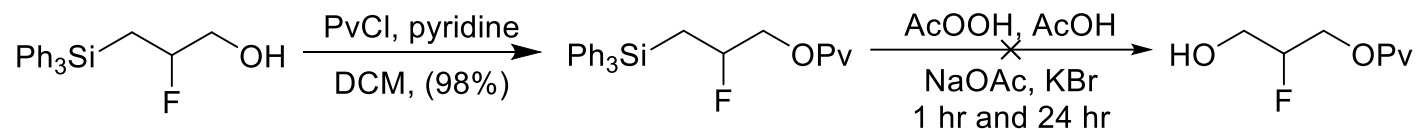
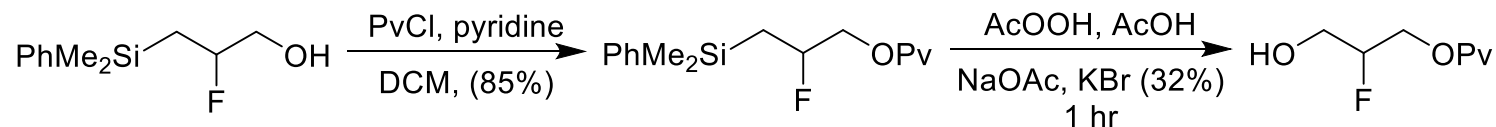
Entry	Conditions	Time
1	$\text{R}_3 = \text{Ph}_3$: TBAF, DCM, Δ	3 hr
2	$\text{R}_3 = \text{Ph}_3$: $\text{BF}_3 \cdot 2\text{AcOH}$, DCM, Δ	5 hr
3	$\text{R}_3 = \text{PhMe}_2$: $\text{BF}_3 \cdot 2\text{AcOH}$, DCM, rt	5 min

- Using a different allylsilane as a masked hydroxy group to oxidize silicon.



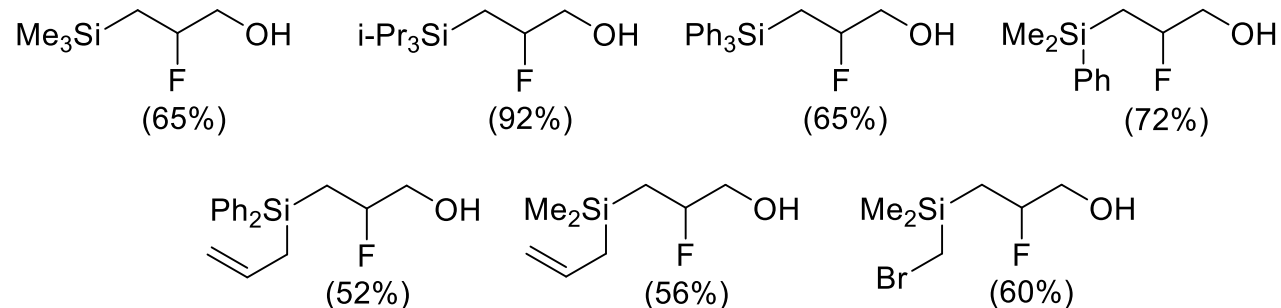
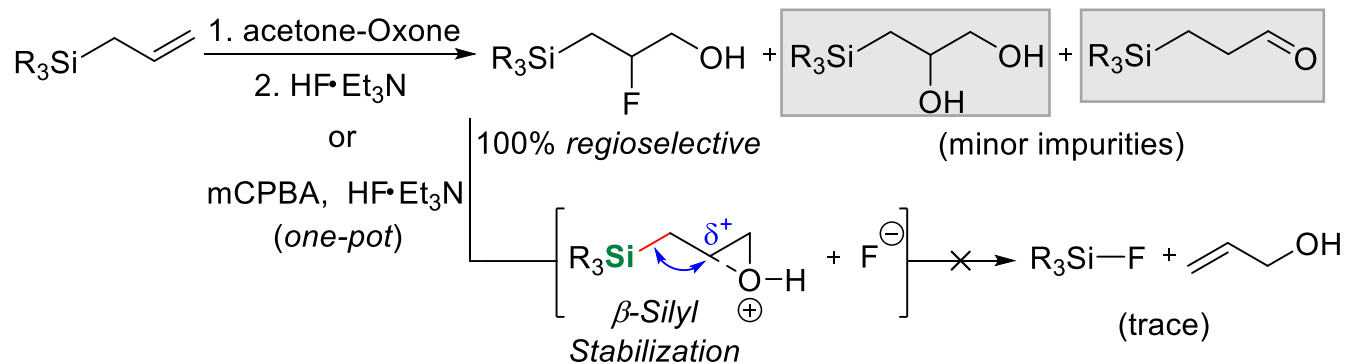
Entry	Conditions	Time
1	BF ₃ ·2AcOH, DCM	30 min
2	I ₂ , MeOH, DCM	1 h
3	I ₂ , CyOH, DCM	1 h

- With our options for a Tamao oxidation becoming limited, we decided to try a one-pot procedure.
- Thankfully this one-pot worked, although in low yield.
- Triphenyl resulted in recovery of starting material.

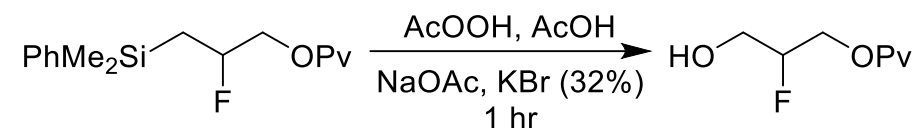


-
- Reaction scheme showing the synthesis of an allyl ether from a silyl-protected fluorinated alcohol.
- Starting material: $\text{PhMe}_2\text{Si}(\delta^+)-\text{CH}_2-\text{CH}(\text{F}, \delta^-)-\text{CH}_2-\text{OPv}$
- Reaction conditions: AcOOH, AcOH followed by $\text{NaOAc, KBr (32\%)} \text{ 1 hr}$
- Intermediate: $\text{HO}-\text{CH}_2-\text{CH}(\text{F})-\text{CH}_2-\text{OPv}$
- Final product: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OPv}$
- Mechanism details:
- The starting material reacts with AcOOH to form a silyl ester intermediate: $\text{Me}_2\text{Si}(\text{OAc})-\text{CH}_2-\text{CH}(\text{F})-\text{CH}_2-\text{OPv}$.
 - The intermediate undergoes a [2,3]-Wittig rearrangement via a cyclic transition state (shown in brackets).
 - The transition state involves the migration of the OPv group from the carbon to the silicon atom, forming a cyclic siloxane intermediate.
 - The final product is the allyl ether: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OPv}$.

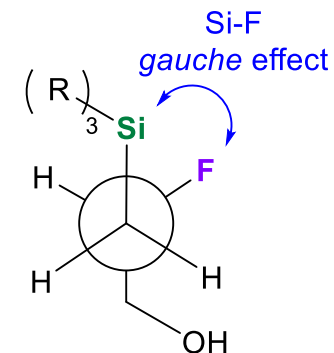
Scope



Tamao Oxidation



First report of a silicon-fluorine *gauche* effect



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Funding:

